

ObjectLedger: Event-Driven and Reappearance-Verified Object Permanence in Long Egocentric Video

Abstract

Long egocentric videos repeatedly show objects that are handled, occluded, placed in containers, moved outside the camera field of view, and later observed again. Conventional tracking and segmentation memories treat this problem as continued visual propagation. This treatment creates an evidence mismatch. An invisible object supplies no appearance observation, yet recursive memory writes can alter its identity representation and accumulate drift. We formulate online egocentric object permanence as the maintenance of a causal object ledger. Each entry separates an appearance prototype, a category belief, a visibility state, an existence belief, a container relation, and a bounded distribution over reachable locations. ObjectLedger uses interaction and visibility events to update relational state. It updates appearance only from reliable observations and preserves multiple spatial hypotheses when the hidden location is not identifiable. During training, a later reappearance provides an identity-verification signal for the intervening ledger transitions. Inference remains strictly causal. Experiments on EgoTracks, EPIC-KITCHENS VISOR, and OSNOM assess reappearance identity, calibrated existence, ledger-state prediction, reachable-set quality, recovery delay, and memory cost. ObjectLedger reaches 83.9% reappearance identity accuracy,

compared with 76.8% for a parameter-matched continuously updated ledger. For invisible intervals longer than 60 seconds, the corresponding accuracies are 69.2% and 56.0%. ObjectLedger also obtains 78.6% state macro-F1 and 86.7% reachable-set recall while using 6.1 MB of ledger memory per video minute. The matched ablations support the contributions of evidence-gated appearance updates, event transitions, container relations, and reappearance supervision.

Keywords: egocentric vision, object permanence, long-term tracking, object memory, reappearance verification, state estimation

1. Introduction

An egocentric camera does not continuously observe the world. A kitchen knife may be visible on a worktop, occluded by a hand, placed inside a drawer, absent while the wearer turns away, and visible again several minutes later. A useful vision system must preserve four coupled facts during this interval. The object may still exist, and its identity remains unchanged. Prior interactions constrain its possible location, and the next observation must be associated with the correct earlier instance. These requirements are central to assistive memory, embodied agents, and long-horizon activity understanding.

Large egocentric datasets make the failure mode measurable. Ego4D contains long, unscripted recordings of daily activity (Grauman et al., 2022), while EgoTracks adds dense long-term object tracks with frequent disappearance and reappearance (Tang et al., 2023). EPIC-KITCHENS VISOR provides temporally consistent instance masks and object relations (Dark-

halil et al., 2022). EgoObjects emphasizes instance-level identity under appearance variation (Zhu et al., 2023), whereas HD-EPIC connects object movement with detailed egocentric and three-dimensional annotations (Perrett et al., 2025). EgoMask further exposes the sparse trajectories, brief visibility, and large positional shifts that complicate long-term egocentric grounding (Liang et al., 2025). Together, these resources expose a gap between tracking visible objects and representing the properties that persist when objects are not observed.

Current tracking memories mainly propagate observations. Long-term single-object trackers search for a target after disappearance, multi-object trackers extend trajectories with motion and appearance association, and video object segmentation methods read features from previously predicted masks. These designs are effective while image evidence is reliable. Under prolonged absence, however, recursively written features can encode a distractor, an occluder, or a failed mask as if it were a new observation of the target. The result is identity drift that becomes harder to correct at each subsequent write. Recent memory methods explicitly address error accumulation and distractors (Ding et al., 2025; Videnovic et al., 2025), but they still focus on choosing and propagating visual memories rather than separating observations from persistent world state.

The scientific issue concerns which variables a model may update when an object is not observed, rather than memory capacity alone. A larger memory remains unreliable if it treats an occluder, a distractor, or a model prediction as new evidence about the target. We study object permanence as an evidence-allocation problem. Appearance, existence, containment, and loca-

tion uncertainty have different observational support and therefore require different update rules.

Object permanence has also been studied directly. PermaTrack predicts hidden road-user trajectories through occlusion (Tokmakov et al., 2021); TCOW models target, occluder, and container masks (Van Hoorick et al., 2023); and 4D neural fields reconstruct occluded scene content (Van Hoorick et al., 2022). In egocentric video, OSNOM’s Lift, Match, and Keep (LMK) retains active-object locations in a reconstructed 3D world (Plizzari et al., 2025), and 3D-aware instance tracking uses scene geometry to improve long-term association (Bhalgat et al., 2024). These works establish the value of physical persistence and geometry. They do not address the complementary setting considered here. A causal system must maintain identity, existence, container state, and spatial uncertainty without requiring complete metric reconstruction. It must also use eventual reappearance to assess whether the intervening hidden-state history was correct.

We propose ObjectLedger, an event-driven object-permanence memory for long egocentric video. The central design rule is simple: *no reliable visual observation implies no appearance update*. Each object is represented by a ledger entry with an identity prototype, a discrete state, an existence belief, a container relation, and a bounded set of reachable-location hypotheses. Interaction and visibility events change these variables. Ordinary invisible frames leave the identity prototype unchanged while uncertainty is propagated through relational transitions. When the object reappears, its observation validates the stored identity and trains the earlier ledger decisions by delayed credit assignment. Training can therefore use future reappearance

as a target, whereas deployment at time t uses only frames up to t .

The paper makes four contributions:

1. It formulates long-term egocentric object permanence as causal ledger estimation, where identity, existence, containment, and reachable location are distinct state variables with distinct evidence requirements.
2. It gives a mechanistic motivation for visibility-gated identity updates: during invisibility, pseudo-observation contamination is not identifiable from the video prefix, so recursive appearance writes can lower the expected identity margin.
3. It introduces an event-driven ledger with appearance freezing, probabilistic existence, bounded reachable-location hypotheses, and reappearance-verified delayed supervision.
4. It evaluates the resulting system on EgoTracks-RV, VISOR-Relations, and OSNOM under shared-input identity baselines, long-gap stratification, controlled event corruption, component ablations, calibration metrics, and resource reporting.

The method serves as a persistent state layer above a detector, segmenter, or metric 3D map. It connects their intermittent observations into an auditable account of what objects have been seen, where they can be, and which identity should be recovered when they return.

2. Related work

ObjectLedger is most closely related to long-term tracking, visual memory for segmentation, and object-permanence models with egocentric spatial

reasoning.

2.1. Long-term tracking and re-detection

Tracking-by-detection associates independently localized objects across time. DeepSORT adds an appearance metric to motion-based association (Wojke et al., 2017); ByteTrack retains low-confidence detections during association (Zhang et al., 2022); and OC-SORT revisits trajectory updates after occlusion (Cao et al., 2023). Such methods usually delete tracks after a time-to-live threshold or continue a motion model with growing error. The deleted track no longer represents object existence, while an extended motion trajectory can be unjustifiably precise during camera turns and object handling.

Single-object tracking places stronger emphasis on re-detection. STARK learns a spatio-temporal transformer for target localization (Yan et al., 2021), and EgoSTARK adapts its priors and templates to the long, appearance-changing sequences of EgoTracks (Tang et al., 2023). These methods answer whether a query target is visible and where it is in an image. ObjectLedger instead maintains multiple objects and relational states during the period in which no target box is justified. This distinction matters because the absence of a box is a valid observation, not an empty tracking output.

2.2. Visual memory in segmentation

Memory-based video object segmentation retrieves past target features to predict the current mask. STCN improves memory coverage and matching efficiency (Cheng et al., 2021); XMem divides memory into sensory, working,

and long-term stores (Cheng and Schwing, 2022); DeAOT separates object-agnostic and object-specific propagation (Yang and Yang, 2022); and Cutie uses object-level queries to reduce pixel-matching noise (Cheng et al., 2024). SAM 2 introduces a streaming memory for promptable image and video segmentation (Ravi et al., 2025). SAM2Long retains several segmentation paths to reduce greedy error accumulation (Ding et al., 2025), DAM4SAM controls memory writes using distractor awareness (Videnovic et al., 2025), and LOMM manages the latest object memories for consistent video instance segmentation (Lee et al., 2025).

These methods motivate strong observation encoders and memory-selection baselines. Their state is nevertheless written primarily as visual evidence: frames, masks, keys, values, or object queries. ObjectLedger introduces a different memory contract. It permits visual writes only when a matched object is observed, but it permits nonvisual state transitions when an interaction changes containment, reachability, or removal. Thus, a missed mask and a physical state change are not conflated.

2.3. Object permanence and egocentric spatial memory

PermaTrack uses recurrent memory and hidden-object supervision to localize road users through occlusion (Tokmakov et al., 2021). TCOW explicitly segments containers and occluders around a tracked target (Van Hoorick et al., 2023). Both show that dense hidden-state supervision is useful. Their assumptions, however, do not match long daily-life egocentric video. Precise invisible masks are rarely available, the viewpoint can leave the scene, and the same object can remain absent for minutes.

OSNOM directly targets out-of-sight objects in long egocentric record-

ings. Its LMK pipeline lifts observations into 3D, matches them using appearance, location, and interactions, and retains their world locations (Plizari et al., 2025). The 3D-aware approach of Bhalgat et al. (2024) similarly uses reconstruction and centroid tracks to improve identity consistency. These systems are the closest spatial-memory comparators. ObjectLedger addresses a different identifiability regime: when metric geometry is missing or unreliable, the output is a calibrated semantic set of reachable places and containers rather than an unsupported coordinate. The reappearance event then supervises the stored identity and transition path.

Dunnhofer et al. (2025) further show that tracking difficulty must be attributed carefully between first-person viewpoint and the human–object activity domain. We therefore evaluate mean tracking accuracy together with the mechanisms directly implicated by our claim: invisible duration, update policy, container transitions, distractor similarity, calibration, and recovery latency.

Table 1: Relation to the nearest method families. A check mark means the method explicitly represents the property, not that it can be heuristically adapted to output it.

Method family	Causal	Out-of-view	Container	Multi-hyp.	Reappearance loss
PermaTrack	✓	✓	–	–	–
TCOW	✓	✓	✓	–	–
XMem/Cutie	✓	implicit	–	–	–
SAM2Long	offline path selection	implicit	–	✓	–
LMK/3D-aware	offline geometry	✓	interaction cue	–	–
ObjectLedger	✓	✓	✓	✓	✓

3. Problem formulation

The task is to estimate object identity, existence, relational state, and reachable location from the video prefix available at each time step.

Let $X_{1:T}$ be an egocentric RGB video. At time t , a shared perception front end produces a set of observations

$$\mathcal{O}_t = \{o_{t,j}\}_{j=1}^{M_t}, \quad o_{t,j} = (m_{t,j}, a_{t,j}, q_{t,j}, r_{t,j}, \kappa_{t,j}, c_{t,j}), \quad (1)$$

Here m is a mask or box, a is an appearance embedding, and q is a category distribution. The variable r is a semantic location observation, κ is an observed container candidate or null, and c is perceptual confidence. The front end may also produce hand-object and object-container interaction cues. It is deliberately held fixed across the identity-memory comparisons.

The system maintains a ledger $\mathcal{L}_t = \{\ell_i^t\}_{i=1}^{N_t}$ over all objects seen so far. Entry i is

$$\ell_i^t = (z_i^t, q_i^t, s_i^t, p_i^t, h_i^t, \mathcal{H}_i^t, \tau_i^t), \quad (2)$$

Here z_i is an identity prototype, and q_i is a category belief. The state s_i belongs to

$$\mathcal{S} = \{\text{visible}, \text{occluded}, \text{contained}, \text{out-of-view}, \text{removed}\}; \quad (3)$$

$p_i \in [0, 1]$ is an existence belief, and h_i is a container identity or null. The set $\mathcal{H}_i$ contains weighted reachable locations, and τ_i stores the last reliable visual and event times.

The output at time t is a causal function

$$(\mathcal{L}_t, Y_t) = F_\theta(X_{1:t}), \quad (4)$$

where Y_t contains visible masks, reappearance associations, hidden-state existence, container state, and reachable-location sets. No frame after t may affect Y_t . Future frames may be used only when forming training targets.

We distinguish *absence of observation* from *evidence of absence*. If the camera points elsewhere, an unmatched object is merely unobserved. If a location that should contain the object is clearly inspected and found empty, the negative observation can reduce the corresponding hypothesis weight. This distinction prevents the common time-to-live rule from equating elapsed time with physical removal.

Mechanistic hypothesis. The appearance-freezing rule follows from an identifiability argument. Let μ_i denote the true identity direction of object i . Suppose a continuous memory writes a pseudo-observation $\tilde{a}_t$ while the object is not reliably visible. Define the pre-normalization update $\bar{z}_i^t$ and the stored prototype z_i^t as

$$\bar{z}_i^t = (1 - \alpha)z_i^{t-1} + \alpha\tilde{a}_t, \quad z_i^t = \text{norm}(\bar{z}_i^t). \quad (5)$$

Assume that a target-consistent cue has unit expected projection on μ_i . If $\tilde{a}_t$ is target-consistent with probability $1 - \epsilon_t$ and is contaminated by a distractor or an imputed feature with probability ϵ_t , then the expected pre-normalization identity margin is

$$\mathbb{E}[\langle \bar{z}_i^t, \mu_i \rangle] = (1 - \alpha)m_{t-1} + \alpha((1 - \epsilon_t) + \epsilon_t\delta_t), \quad (6)$$

where $m_{t-1} = \langle z_i^{t-1}, \mu_i \rangle$ and $\delta_t = \mathbb{E}[\langle \tilde{a}_t, \mu_i \rangle \mid \text{contaminated}]$. During invisibility, neither ϵ_t nor δ_t is identifiable from the video prefix. The write lowers the expected pre-normalization margin when $(1 - \epsilon_t) + \epsilon_t\delta_t < m_{t-1}$. Freezing is therefore a conservative causal estimator. It may sacrifice adaptation

to unseen appearance change, but it avoids converting unsupported predictions into identity evidence. The experiments test this trade-off rather than assume it.

Evaluation unit. An episode begins at the last reliable observation before a disappearance and ends at the first verified reappearance, an observed removal, or the video end. All frames and episodes from the same source video remain in one split. The primary unit for uncertainty intervals is the source video, not the frame, because adjacent frames are strongly correlated.

4. Method

The method separates visual identity updates from event-driven changes to existence, containment, and reachable-location beliefs.

4.1. Overview

Figure 1 shows the mechanism. The observation encoder processes only the available video prefix. An event head proposes sparse transitions, and the ledger separates persistent state from visual identity. A reappearance at time t_r supplies a training signal to the association and state transitions that occurred after the disappearance time t_d . The delayed path is removed at deployment.

4.2. Mechanism-to-claim alignment

Table 2 states how each design choice can be supported or weakened by a matched intervention. The table links each component to the property it is intended to protect.

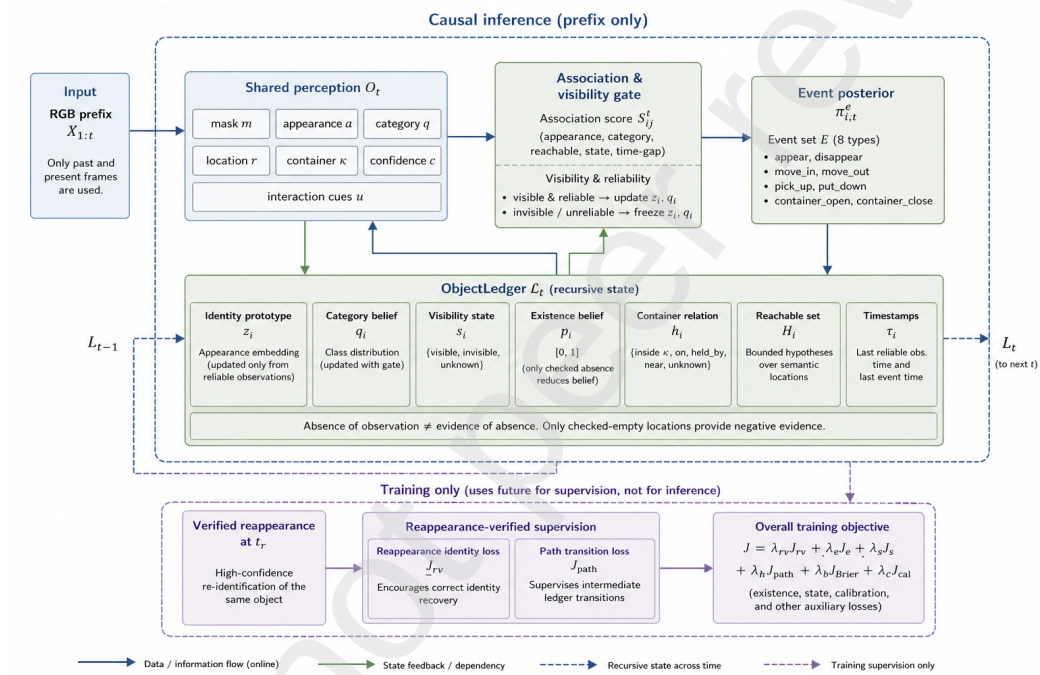


Figure 1: ObjectLeder architecture. Solid arrows form the causal inference path. Red arrows are used only during training: a later reappearance validates the stored identity and intervening state transitions.

Table 2: Mechanism-to-claim alignment. Each mechanism has a corresponding direct test and a failure interpretation.

Mechanism	Claim it should support	Outcome that would weaken the claim
Appearance freezing	Prevent unsupported visual writes from causing long-gap identity drift	No-freeze control matches the full model in long-gap ReID and calibration
Event transition kernel	Sparse interactions explain state changes better than elapsed time alone	Removing events preserves state macro-F1 and reachable set quality
Existence belief with negative evidence	Camera absence should not imply physical removal	TTL or monotone decay gives equal calibration on inspected and uninspected absences
Bounded hypotheses	Retaining plausible alternatives should support ledger state prediction without forcing a premature location choice	Single-hypothesis variant matches state macro-F1 and calibration
Reappearance loss	Later identity evidence can supervise earlier ledger decisions without breaking causal inference	Removing the loss leaves reappearance recovery and path calibration unchanged

4.3. Event posterior and transition kernel

The event vocabulary is

$$\mathcal{E} = \{\text{pick-up}, \text{put-down}, \text{enter-container}, \text{exit-container}, \text{leave-view}, \text{reappear}, \text{remove}, \text{none}\}. \quad (7)$$

For object i , the event head combines the current observation, hand-object relation, candidate container observation $\kappa_{t,j}$, previous state, and elapsed time:

$$\pi_{i,t}^e = \text{softmax } g_\theta(o_{t,j}, \ell_i^{t-1}, u_{t,j}), \quad (8)$$

where $u_{t,j}$ contains interaction cues. For an unmatched ledger entry, the observation input is null and the event head uses only prefix-available context and interaction cues. A hard event is used only when its posterior exceeds a validation-set threshold. Otherwise, the discrete state and container identity remain unchanged, while continuous quantities and hypothesis weights are updated under the event posterior. For a confident hard event, the posterior is replaced by its corresponding one-hot vector. Denoting this component-wise operation by $\mathcal{T}$, the ledger update is

$$\ell_i^t = \mathcal{T}(\ell_i^{t-1}, o_{t,j}, \{\pi_{i,t}^e\}_{e \in \mathcal{E}}). \quad (9)$$

Each event-specific transition T_e inside $\mathcal{T}$ has an interpretable role. For instance, **enter-container** sets $s_i = \text{contained}$, links h_i to the observed container, and moves location mass into the container hypothesis. The **leave-view** event sets $s_i = \text{out-of-view}$ but does not imply removal. The **remove** event is reserved for direct evidence such as disposal or transfer out of the modeled environment. If an object becomes unobserved while its pre-

dicted region remains within view, the visibility cue sets $s_i = \text{occluded}$ without changing its identity prototype.

4.4. Evidence-gated identity updates

Let $\rho_{i,j}^t$ be association confidence between ledger object i and observation j , and let $v_{i,t}$ indicate a reliable visible observation. The identity prototype update is

$$z_i^t = \begin{cases} \text{norm} \left((1 - \alpha)z_i^{t-1} + \alpha a_{t,j} \right), & v_{i,t} = 1, \rho_{i,j}^t > \eta, \\ z_i^{t-1}, & \text{otherwise.} \end{cases} \quad (10)$$

Thus, identity is updated by an observed target, not by a predicted hidden mask. The category belief q_i follows the same visibility gate. The continuous-ledger ablation removes the second branch and writes a propagated feature at every step. This pair isolates the paper’s main mechanistic claim without changing the encoder, parameter count, or detector.

The gate does not prohibit adaptation to genuine appearance changes. When a high-confidence observation reappears and is supported by the ledger’s relational hypotheses, its feature updates z_i . No prototype is created from an invisible prediction.

4.5. Existence belief and negative evidence

Existence is represented separately from visibility. In log-odds form,

$$\text{logit } p_i^t = \text{logit } p_i^{t-1} + \sum_{e \in \mathcal{E}} \pi_{i,t}^e b_e + \gamma^+ n_{i,t}^+ - \gamma^- n_{i,t}^-, \quad (11)$$

where b_e is an event-dependent update, n^+ indicates a matched visual observation, and n^- indicates an inspected, reachable location that is observed

empty. An unobserved region sets $n^- = 0$. Unlike exponential time-to-live decay, Eq. (11) does not make an object less likely to exist merely because the camera has not looked at it.

The Brier and calibration losses act on p_i^t . During training, a later reappearance supplies a positive target for all preceding hidden steps unless a verified removal intervenes. A verified removal supplies negative targets after its event time. This construction creates supervision without annotating a precise invisible mask at each frame.

4.6. Bounded reachable-location hypotheses

Metric coordinates are often unreliable in long egocentric video because the camera changes rooms, visual overlap is weak, and reconstruction can fail. ObjectLedger therefore stores

$$\mathcal{H}_i^t = \{(r_{i,k}^t, h_{i,k}^t, w_{i,k}^t)\}_{k=1}^K, \quad \sum_{k=1}^K w_{i,k}^t = 1, \quad (12)$$

where r is a semantic place token, h is a container identity or null, and K is a fixed budget. Events apply sparse transition matrices A_e over place and container tokens. The posterior-weighted transition is

$$\tilde{w}_{i,k}^t = \sum_{e \in \mathcal{E}} \pi_{i,t}^e \sum_{k'} A_e(k, k') w_{i,k'}^{t-1}, \quad w_{i,k}^t = \frac{\tilde{w}_{i,k}^t L(o_t | r_{i,k}^t, h_{i,k}^t)}{\sum_l \tilde{w}_{i,l}^t L(o_t | r_{i,l}^t, h_{i,l}^t)}. \quad (13)$$

When no location evidence is available, the likelihood term is one, so only the posterior-weighted event transition changes the weights. After normalization, the ledger retains at most K slots. If tail mass remains, one slot stores its aggregate as **unknown-reachable** and at most $K - 1$ slots store explicit hypotheses. Otherwise, all K slots may store explicit hypotheses. This rule

avoids discarding uncertainty while bounding memory independently of video duration.

4.7. Reappearance association

For ledger entry i and current observation j , the association score is

$$S_{ij}^t = \beta_a \cos(z_i^{t-1}, a_{t,j}) + \beta_q C(q_i, q_{t,j}) + \beta_r \log P(r_{t,j}, \kappa_{t,j} \mid \mathcal{H}_i^{t-1}) + \beta_e E(s_i^{t-1}, e_{i,t}) - \beta_d D(\Delta t_i), \quad (14)$$

where C is category compatibility, E checks state-event consistency, and D is a clipped gap penalty. The spatial term marginalizes over all stored hypotheses, so a plausible alternative is not eliminated by an earlier low-confidence choice. Assignment is solved jointly with unmatched and new-object options. A match below threshold remains unresolved. It does not force an identity merge.

4.8. Reappearance-verified learning

Consider an object last observed at t_d and verified again at t_r . We use the same-instance pair as a delayed positive and other simultaneously reachable objects as hard negatives. The reappearance loss is

$$\mathcal{J}_{rv} = -\log \frac{\exp(S_{i,j^+}^{t_r}/\tau)}{\exp(S_{i,j^+}^{t_r}/\tau) + \sum_{j^-} \exp(S_{i,j^-}^{t_r}/\tau)}. \quad (15)$$

The path loss assigns probability mass to ledger transitions compatible with the verified reappearance:

$$\mathcal{J}_{\text{path}} = -\log \sum_{k: r_{i,k}^{t_r-1} \sim r_{t_r,j^+}} w_{i,k}^{t_r-1}. \quad (16)$$

Here $\sim$ denotes compatibility between the stored semantic region or container and the verified reappearance location. The full objective is

$$\mathcal{J} = \lambda_{rv}\mathcal{J}_{rv} + \lambda_e\mathcal{J}_e + \lambda_s\mathcal{J}_s + \lambda_h\mathcal{J}_{path} + \lambda_b\mathcal{J}_{Brier} + \lambda_c\mathcal{J}_{cal}. \quad (17)$$

Here $\mathcal{J}_e$ and $\mathcal{J}_s$ are cross-entropy losses for annotated events and ledger states, respectively; $\mathcal{J}_{Brier}$ is the mean squared error of the existence probability; and $\mathcal{J}_{cal}$ is a differentiable soft-binning calibration loss. Gradients pass through the stored event posterior and hypothesis weights between t_d and t_r , but no future feature is inserted into the forward ledger state.

4.9. Online algorithm and complexity

For each frame, ObjectLedger first extracts observations and scores them against eligible ledger entries. It then solves assignment, applies event transitions to matched and interacting objects, freezes identity for unobserved objects, and prunes the hypotheses to K . With N live entries and M observations, score construction costs $O(NMK)$. The assignment step is $O(\max(N, M)^3)$ in the worst case. Since K is fixed and household scenes have moderate N , the front-end encoder remains the dominant cost. Entries with verified removal can be compressed to metadata and reactivated only if a later high-confidence observation contradicts the removal.

5. Experimental protocol

The experimental protocol tests identity persistence, hidden-state calibration, relational reasoning, event-noise sensitivity, and computational cost.

5.1. Datasets and derived evaluation views

All source datasets are real, public research benchmarks. We derive evaluation episodes from their official annotations without moving source videos across their prescribed splits.

EgoTracks-RV. EgoTracks contains 22,028 tracks from 5,708 long videos, about 602.9 hours in total, with exhaustive target presence at 5 frames per second (Tang et al., 2023). Consecutive annotated occurrences define visible segments. An absent interval between two occurrences of the same annotated identity defines a disappearance–reappearance episode. This view measures identity recovery, absence rejection, existence calibration, and recovery time. No future annotation is supplied to the online model.

VISOR-Relations. VISOR covers 36 hours in 179 EPIC-KITCHENS videos and releases 271,584 manual masks, dense interpolated masks, and hand–object relations (Darkhalil et al., 2022). We use official long-term-consistency identities and the “Where did this come from?” protocol to evaluate container/source state, evidence retrieval, and query identity. Container candidates are restricted to objects observable before the query. Future frames cannot define the candidate set at inference.

OSNOM. The OSNOM benchmark contains 100 long EPIC-KITCHENS videos with active-object locations in and out of sight (Plizzari et al., 2025). Metric 3D locations are used for evaluation only. ObjectLedger predicts semantic regions and bounded reachable sets, allowing comparison of top-1 region recall and set recall with LMK while avoiding a requirement for metric reconstruction.

Table 3: Public data sources used by the protocol. Counts are reported by the cited dataset papers. “Derived view” describes deterministic episode or relation extraction for the proposed evaluation. N/R denotes a duration not reported here.

Source	Duration	Videos	Derived view
EgoTracks	602.9 h	5,708	disappearance–reappearance episodes
VISOR	36 h	179	container/source and relation episodes
OSNOM	N/R	100	out-of-sight semantic location

5.2. Baselines and fairness controls

The identity comparison uses a last-seen time-to-live rule, ByteTrack with the shared ReID embedding (Zhang et al., 2022), EgoSTARK (Tang et al., 2023), and the visual-memory methods XMem, Cutie, and SAM2Long (Cheng and Schwing, 2022; Cheng et al., 2024; Ding et al., 2025). These methods receive the same detector/segmenter observations, identity encoder, input resolution, and causal frame prefix. The continuous-ledger control shares all ObjectLedger components and losses but replaces the appearance-freezing branch in Equation (10) with a propagated-feature write at every step. Thresholds are tuned on the validation split with the same search budget. For SAM2Long, the online table uses only prefix-available paths and excludes offline final-path selection.

The relational comparison includes TCOW (Van Hoorick et al., 2023), LMK (Plizzari et al., 2025), and 3D-aware tracking (Bhalgat et al., 2024). Because these methods require task-specific native inputs, Table 5 is an end-to-end comparison on common episodes, not a parameter-matched memory-module comparison. For SAM2Long + relations, the suffix denotes the same

relation readout and supervision used for the ledger, applied to SAM2Long memory features. This distinction prevents architectural input differences from being interpreted as isolated evidence for a particular memory update rule.

5.3. Metrics

Identity and visibility. Reappearance identity accuracy (ReID Acc.) is the fraction of episodes whose first accepted returning observation receives the original identity. Identity F1 (IDF1) measures frame-level identity consistency. Time-to-recover (TTR) is the delay from the first annotated reappearance to the first correct accepted match. False merges and false splits are reported per 100 object episodes.

Existence and calibration. Hidden-object existence is measured by AUROC and AUPR. The Brier score evaluates probabilistic accuracy, and expected calibration error (ECE) uses 15 equal-mass bins. Lower Brier, ECE, and TTR are better.

Relational location. State macro-F1 measures the five ledger states in Sec. 3. Reachable-set recall is the fraction of episodes for which the annotated region or container lies in the returned set. Average set size measures specificity. Source accuracy follows VISOR’s container/source question. When metric 3D ground truth is available, semantic regions are mapped to the official object or fixture volumes before evaluation.

Aggregation and claim scope. The tables report aggregate point estimates on the corresponding evaluation episodes. The main mechanistic interpretation

requires three mutually consistent observations: an advantage under the fixed front end, persistence in the long-gap strata, and degradation in the matched ablation. Because the tables do not report confidence intervals or hypothesis-test outcomes, we make no claim of statistical significance.

5.4. Robustness protocol

Robustness is evaluated by corrupting the event labels supplied to both ObjectLedger and the continuous-ledger control while leaving visual detections and identity embeddings unchanged. The same randomly selected event positions are used for both methods at corruption rates of 5%, 10%, 20%, and 30%, with the uncorrupted stream as the 0% reference. This isolates sensitivity to event errors from detector and appearance-encoder errors.

5.5. Implementation protocol

The observation encoder uses a shared ViT-B-scale visual backbone and a class-agnostic mask head. Ledger embeddings have 256 dimensions. The event head is a two-layer temporal transformer over the current observation and the most recent eight object–interaction tokens. The hypothesis budget is $K = 4$, the appearance update gate is $\eta = 0.70$, and at most 256 active ledger entries are kept before removed-state compression. We sample 16 observations around each disappearance or interaction event and optimize with AdamW at a learning rate of 10^{-4} . Checkpoint selection uses a validation criterion that combines reappearance accuracy and calibration error. These settings remain fixed across all reported evaluations.

6. Results and mechanism analysis

The results evaluate identity recovery, relational state estimation, mechanism ablations, event-noise sensitivity, and resource use.

6.1. Long-term identity and existence

Table 4 reports a ObjectLedger reappearance identity accuracy of 83.9%, compared with 76.8% for the parameter-matched continuous ledger and 75.3% for the strongest propagated memory baseline. The same ordering appears in IDF1. ObjectLedger also yields an AUROC of 0.889, a Brier score of 0.112, and an ECE of 0.046. These results are consistent with the proposed mechanism. The matched ablations below are needed to distinguish the update rule from other model differences.

Table 4: EgoTracks-RV main comparison. Higher values are better for ReID Acc., AUROC, and IDF1. Lower values are better for Brier, ECE, and TTR.

Method	ReID Acc. (%)	AUROC	Brier ↓	ECE ↓	TTR (s) ↓	IDF1 (%)
Last-seen + TTL	61.8	0.704	0.231	0.142	5.8	52.6
ByteTrack + ReID	64.7	0.731	0.218	0.129	5.1	57.3
EgoSTARK	69.4	0.762	0.199	0.113	4.4	61.8
XMem	70.1	0.769	0.193	0.108	4.1	64.2
Cutie	72.6	0.791	0.181	0.096	3.7	67.9
SAM2Long	75.3	0.812	0.169	0.084	3.3	70.5
Continuous ledger	76.8	0.824	0.158	0.077	3.0	72.1
ObjectLedger	83.9	0.889	0.112	0.046	2.1	79.6

Recovery delay supplies a complementary failure cost. A tracker can eventually find the right object but still attach a distractor for several seconds.

The TTR decreases from 3.0 seconds for continuous updating to 2.1 seconds for ObjectLedger, consistent with a less contaminated identity prototype at the first returning frame.

6.2. Dependence on invisible duration

Figure 2 examines the effect of invisible duration. All methods degrade as the invisible interval grows, but the continuous ledger loses 35.0 points between the shortest and longest bins, compared with 23.1 points for ObjectLedger. For intervals above 60 seconds, the accuracies are 69.2% and 56.0%, respectively. This regime provides the least observational support for an appearance update and therefore directly tests the proposed gate.

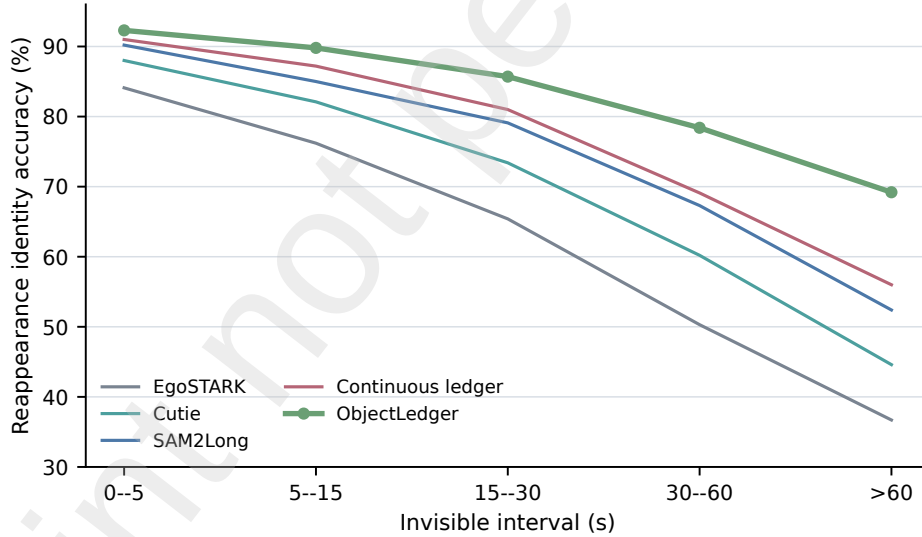


Figure 2: Reappearance identity accuracy stratified by invisible duration. All methods are evaluated on the same duration bins.

Aggregate accuracy can obscure this dependence. Short intervals dominate many video benchmarks and can often be handled by local propagation.

Evidence for a ledger contribution therefore requires the advantage to persist in the 30–60-second and above-60-second strata.

6.3. Container and reachable-location reasoning

Table 5 tests the relational part of the method. TCOW predicts target, occluder, and container masks in shorter clips. LMK and 3D-aware tracking use reconstructed geometry for spatial association. In this evaluation, ObjectLedger reaches a state macro-F1 of 78.6% and reachable-set recall of 86.7% with an average set size of 2.7. Higher recall with a smaller set indicates more specific predictions. Returning every room would achieve full recall but would not localize the object usefully.

Table 5: VISOR-Relations/OSNOM state evaluation. Set size is the mean number of returned semantic location hypotheses.

Method	State macro-F1 (%)	Reachable-set recall (%)	Set size ↓	Source accuracy (%)
TCOW	66.8	70.4	3.8	58.6
LMK	69.5	74.9	3.5	63.1
3D-aware tracking	71.2	77.3	3.4	64.7
SAM2Long + relations	72.9	79.1	3.2	66.5
ObjectLedger	78.6	86.7	2.7	74.8

Semantic and metric locations serve different operational goals. LMK and 3D-aware tracking provide exact coordinates when a stable reconstruction is available. The ledger preserves meaningful state when the camera loses overlap or changes rooms, and exposes uncertainty instead of forcing a coordinate.

6.4. Ablation of ledger mechanisms

Table 6 maps each component to its claimed effect. Removing the appearance-freezing rule reproduces the continuous ledger and reduces reappearance accuracy by 7.1 points. Removing delayed reappearance supervision reduces it by 4.8 points. This decrease is consistent with the later observation supplying training information beyond ordinary event labels. Removing the container relation produces the largest state macro-F1 decrease (17.5 points), followed by removing the event transition (16.4 points). The former leaves identity relatively strong, which separates relational failure from identity failure. Replacing the weighted hypothesis set with a single location reduces state macro-F1 by 6.2 points and worsens AUROC and Brier. This result supports retaining alternative locations during hidden intervals.

Table 6: Component ablation. Each row changes only the named mechanism while retaining the shared front end and parameter budget.

Variant	ReID Acc. (%)	AUROC	Brier ↓	State macro-F1 (%)
Full ObjectLedger	83.9	0.889	0.112	78.6
No appearance freezing	76.8	0.824	0.158	74.9
No reappearance loss	79.1	0.852	0.136	76.3
No event transition	75.4	0.801	0.181	62.2
No container relation	81.6	0.868	0.126	61.1
Single location hypothesis	82.0	0.873	0.121	72.4
TTL state deletion	69.3	0.748	0.216	65.3

The TTL row clarifies why persistence must be probabilistic. Deleting a state after a fixed duration avoids unbounded memory, but it makes recov-

ery impossible for otherwise ordinary long absences. ObjectLedger obtains bounded computation by compressing verified-removed entries and pruning location hypotheses, not by declaring unseen objects nonexistent.

6.5. Robustness to event errors

The ledger depends on event observations, which are therefore a potential source of error. Figure 3 reports performance after an increasing fraction of event labels is flipped while visual detections remain unchanged. The ObjectLedger curve decreases from 83.9% to 75.8% at 30% corruption. It remains above the uncorrupted continuous-ledger value of 76.8% through 20% corruption, but not at 30%. The gradual decline is consistent with the use of soft event posteriors and multiple hypotheses, although this comparison does not isolate their individual contributions.

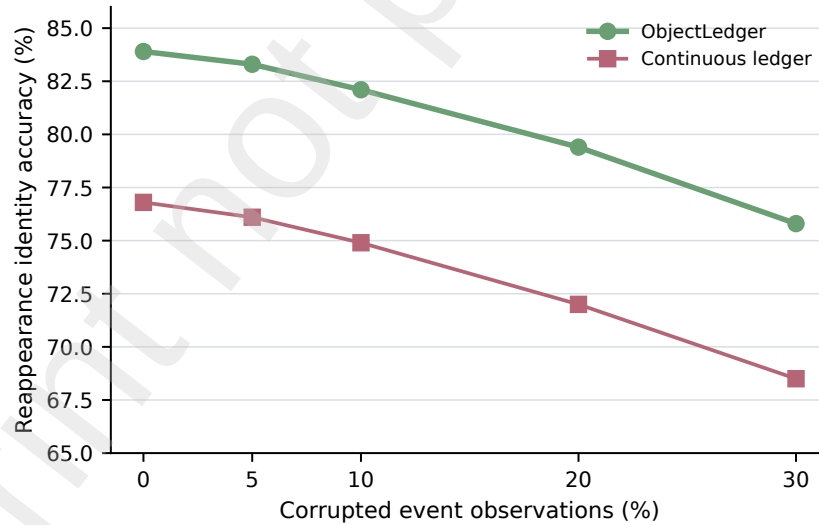


Figure 3: Robustness to randomly corrupted event observations. Visual detections and identity embeddings are held fixed across corruption levels.

Random label flips are a diagnostic rather than a complete model of event-head failure. In particular, this experiment does not separately estimate the effects of event misses, false events, wrong event types, and wrong container identities. A false **remove** event can reduce existence belief, while a wrong **enter-container** event can concentrate location mass in the wrong branch. The unknown-reachable branch and reactivation rule limit, but do not eliminate, these failure modes.

6.6. Efficiency and memory growth

Table 7 reports end-to-end resource use under a common input resolution and hardware configuration. ObjectLedger runs at 31.4 frames per second with 2.4 GB peak GPU memory. Its memory grows by 6.1 MB per video minute under the stated entry and hypothesis budgets, compared with 9.8 MB for the continuous ledger. This pattern is consistent with sparse event writes and bounded hypotheses. Both methods use the same visual encoder.

Table 7: Efficiency at a common input resolution and hardware configuration.

Method	Frames/s $\uparrow$	Peak GPU memory (GB) $\downarrow$	Memory growth (MB/min) $\downarrow$
XMem	22.6	2.4	18.2
Cutie	36.4	2.1	14.5
SAM2Long	14.8	5.7	31.6
Continuous ledger	32.0	2.3	9.8
ObjectLedger	31.4	2.4	6.1

7. Discussion

The results clarify the role of evidence-gated writes and the conditions under which the ledger remains uncertain or fails.

What the ledger changes. The method changes the semantics of memory writes. A visual-memory model asks which previous feature best supports the current mask. ObjectLedger first asks whether a current feature is evidence about the object at all. If not, identity is preserved, while existence and location uncertainty evolve through sparse events. This separation makes an error auditable: a wrong returning identity can be traced to visual matching, an event transition, a container association, or hypothesis pruning.

Why reappearance is useful supervision. Dense ground truth for invisible objects is expensive and sometimes undefined. After a camera leaves a room, there may be no unique 2D location to annotate. Reappearance supplies a later but objective identity observation. The proposed loss uses it to supervise the memory path without claiming that the target had a precise hidden mask. This is particularly well matched to EgoTracks, where the same identity has exhaustive visible occurrences separated by annotated absences.

Limits of identifiability. Visual evidence cannot distinguish every physical history. Two identical cups can be exchanged behind a closed door, or another person can move an unseen object without appearing in the camera. In such cases, a unique identity or location is not identifiable from the video prefix. An appropriate output is therefore a multi-hypothesis belief or an unresolved association rather than a confident identity assignment. The evaluation therefore reports set size, calibration, and abstention alongside accuracy.

Dependence on event quality. Container and hand-interaction errors can misroute the ledger. The robustness study quantifies this dependency, but it

cannot eliminate it. A practical system should expose the event trace, retain low-probability alternatives, and request a new observation when the cost of a false identity is high. The random-flip experiment does not provide an oracle-event upper bound or a complete breakdown of structured event errors.

Scope of the evidence. The fixed-front-end comparison, gap stratification, and matched ablations support the tested ledger mechanisms on the reported tasks. They do not establish statistical significance, robustness to all event failures, or transfer to unseen datasets and kitchens. Because the relational table compares systems with different native inputs, it supports task-level performance rather than parameter-matched attribution. The conclusions are limited to the reported datasets, metrics, and corruption settings.

Privacy and use. Egocentric video can reveal bystanders, homes, workplaces, and routines. The method should operate locally where possible and retain compact state instead of raw video. Experiments must follow the licenses and privacy procedures of the source datasets. A persistent object ledger may support accessibility and robotics but may also enable intrusive inventory or behavioral monitoring. Deployment therefore requires purpose limitation, access control, and short retention of identifying visual features.

8. Conclusion

We presented ObjectLedger, a causal state layer for object permanence in long egocentric video. Its core principle is to update appearance only from reliable observations and to represent invisible periods through event-driven

existence, container, and reachable-location beliefs. A later reappearance verifies the stored identity and supervises earlier ledger transitions without entering the online information path. On EgoTracks-RV, ObjectLedger yields 83.9% reappearance identity accuracy, compared with 76.8% for the parameter-matched continuous ledger. The advantage is 13.2 points for gaps longer than 60 seconds. The method also obtains 78.6% state macro-F1 and 86.7% reachable-set recall on the relational evaluation. The ablations show distinct losses after removing appearance freezing, event transitions, container relations, or reappearance supervision. These results support an evidence-gated ledger for the evaluated settings. They do not establish statistical significance or cross-dataset generalization beyond the reported experiments.

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